

Meaning-specific age-of-acquisition estimates for 562 Chinese disyllabic homonyms

This supplement provides additional methodological details, full participant instructions, and complete statistical tables corresponding to the analyses reported in the main text. Section S1 describes dictionary preprocessing, construction of the Experiment 1 matched benchmark set, and preparation of the meaning-level materials. Section S2 reports the full model outputs cited in the manuscript (Tables S1–S7). All stimuli, anonymized trial-level data, and analysis scripts are available in the public GitHub repository reported in the main article:

<https://github.com/Patrickbigbig/meaning-specific-AoA-of-562-Chinese-homonyms>

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S1. Additional methodological details

S1.1 Dictionary preprocessing and lexical classification

Lexical entries were extracted from the *Modern Chinese Dictionary* (7th ed.; Institute of Linguistics, 2016). Raw dictionary entries were parsed to obtain the headword, part-of-speech (POS) label, sense definitions, and example sentences. Multiple senses within an entry were split into separate meaning-level records. During preprocessing, XML-style markup, hyperlinks, formatting artifacts, and variant-character notices were removed. We retained only linguistically relevant content needed for meaning-level norming: the headword, POS label, each distinct sense definition, and all illustrative examples. Example delimiters such as " | " and numbered markers (e.g., "❶", "❷") were normalized during extraction so that each sense record could be linked to its own example material.

Following the operational definitions used in the main text, we classified disyllabic standard Mandarin word forms into three lexical types according to dictionary structure. **Homonyms** were

disyllabic word forms listed under two or more separate dictionary entries whose entries corresponded to different meanings (and in some cases different pronunciations). **Polysemes** were disyllabic word forms listed under a single entry containing two or more senses. **Monosemes** were disyllabic word forms listed under a single entry with one sense. We excluded multiple-entry cases that reflected pronunciation-only distinctions without a corresponding meaning distinction.

We first restricted the dictionary to disyllabic headwords. From 72,225 total dictionary entries, this yielded 45,311 disyllabic entries. Among these, 819 disyllabic word forms were represented by two separate dictionary entries and were treated as candidate homonyms. We then excluded: (a) variant or non-standard written forms corresponding to a standardized headword, (b) dialect-marked senses, (c) entry splits created only to distinguish pronunciation without a meaning contrast, and (d) senses whose standard written form is now recorded under another headword. After filtering, **562 disyllabic homonymous word forms** remained for the database.

For benchmarking in Experiment 1, only homonyms with the psycholinguistic property records needed for feature-based matching were eligible. This yielded **125 homonymous word forms comprising 333 meanings**. The remaining **437 homonymous word forms comprising 1,108 meanings** were assigned to Experiment 2.

Table A1

Dictionary preprocessing and assignment to experiments

Stage	Unit	Remaining <i>n</i>	Notes
Raw <i>Modern Chinese Dictionary</i> entries	Entries	72,225	Full dictionary before restriction
Disyllabic entries	Entries	45,311	Two-sinogram entries only
Candidate homonyms	Word forms	819	Word forms represented by two separate entries
Final homonym database	Word forms	562	After exclusion of variants, dialect-marked senses, pronunciation-only splits, and senses reassigned to other standard headwords
Experiment 1 homonyms	Word forms / meanings	125 / 333	Eligible for 91-feature matching
Experiment 2 homonyms	Word forms / meanings	437 / 1,108	Remaining homonyms rated online

For all retained homonyms, the released database also includes two additional annotations.

Homograph status indicates whether the dictionary entries for a shared written form differ in pronunciation (including consonant, vowel, or tone differences, and cases involving neutral-tone

realization of the second syllable). **Within-entry polysemy** indicates whether one of the homonymous entries itself contains multiple dictionary senses. In Experiment 1, 60 homographs and 55 homonyms with within-entry polysemy were included; in Experiment 2, the corresponding counts were 153 and 152.

S1.2 Construction of the Experiment 1 benchmark set

S1.2.1 Overview

Experiment 1 was designed as a tightly controlled benchmark for comparing age-of-acquisition ratings across three lexical ambiguity types. The final benchmark set comprised:

- **125 homonymous word forms** contributing **333 meanings**
- **125 polysemous word forms** contributing **333 meanings**
- **333 monosemous word forms** contributing **333 meanings**

Thus, the benchmark set contained **999 meaning-level items** in total.

Table A2

Composition of the Experiment 1 benchmark set

Word type	Word forms	Meanings
Homonyms	125	333
Polysemes	125	333
Monosemes	333	333
Total	583	999

Candidate polysemes and monosemes were drawn from simplified-Chinese psycholinguistic resources that overlapped with our dictionary-derived item pool, including the Chinese Lexical Database (Sun et al., 2018), MELD-SCH (Tsang et al., 2018; Xu & Li, 2020; Xu et al., 2022), and the simplified-Chinese word-level AoA norms of Xu et al. (2021). Matching was anchored on the homonym set because it was the limiting pool.

S1.2.2 Feature space and normalization

Matching was based on a **91-variable feature space** spanning affective ratings, lexical frequency, lexical-decision performance, orthographic structure, phonological statistics, and distributional/character-based measures. To avoid overweighting homonyms that had multiple meaning-level records, standardization was carried out at the **headword level**.

For each feature j , we first deduplicated the dataset so that each headword contributed once to the computation of the mean and standard deviation. Standardized values were then computed as:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

where x_{ij} is the raw value of feature j for headword i , and μ_j and σ_j are the mean and standard deviation of feature j over the deduplicated headword set. The resulting standardized values were then mapped back to all meaning-level records associated with that headword. This ensured that homonyms with multiple meanings did not disproportionately influence scaling of the feature space.

The 91 feature names are reproduced below verbatim from the source datasets.

- | | |
|--|-----------------------------------|
| 1. ValenceArousal_Valence_Mean | 38. CLD_C2FrequencyRawWeibo |
| 2. ValenceArousal_Valence_SD | 39. CLD_PhonologicalFrequency |
| 3. ValenceArousal_Arousal_Mean | 40. CLD_C1PhonologicalFrequency |
| 4. ValenceArousal_Arousal_SD | 41. CLD_C2PhonologicalFrequency |
| 5. ValenceArousal_Men_Valence_Mean | 42. CLD_C1FamilySize |
| 6. ValenceArousal_Men_Valence_SD | 43. CLD_C2FamilySize |
| 7. ValenceArousal_Women_Valence_Mean | 44. CLD_C1FamilyFrequency |
| 8. ValenceArousal_Women_Valence_SD | 45. CLD_C2FamilyFrequency |
| 9. ValenceArousal_Men_Arousal_Mean | 46. CLD_Strokes |
| 10. ValenceArousal_Men_Arousal_SD | 47. CLD_C1Strokes |
| 11. ValenceArousal_Women_Arousal_Mean | 48. CLD_C2Strokes |
| 12. ValenceArousal_Women_Arousal_SD | 49. CLD_C1Pixels |
| 13. SUBTLEXWord_WCount | 50. CLD_C2Pixels |
| 14. SUBTLEXWord_W/million | 51. CLD_C1PictureSize |
| 15. SUBTLEXWord_logW | 52. CLD_C2PictureSize |
| 16. SUBTLEXWord_W-CD | 53. CLD_C1OLDPixels |
| 17. SUBTLEXWord_W-CD% | 54. CLD_C2OLDPixels |
| 18. SUBTLEXWord_logW-CD | 55. CLD_PhonologicalN |
| 19. MELD_RT | 56. CLD_C1PhonologicalN |
| 20. MELD_RTSD | 57. CLD_C2PhonologicalN |
| 21. MELD_zRT | 58. CLD_PLD |
| 22. MELD_zRTSD | 59. CLD_C1PLD |
| 23. MELD_ERR | 60. CLD_C2PLD |
| 24. Concreteness_Mean of Valid Ratings | 61. CLD_MeanPhonemeFrequency |
| 25. Concreteness_SD of Valid Ratings | 62. CLD_C1MeanPhonemeFrequency |
| 26. CLD_Phonemes | 63. CLD_C2MeanPhonemeFrequency |
| 27. CLD_Frequency | 64. CLD_MinPhonemeFrequency |
| 28. CLD_C1Frequency | 65. CLD_C1MinPhonemeFrequency |
| 29. CLD_C2Frequency | 66. CLD_C2MinPhonemeFrequency |
| 30. CLD_FrequencyRaw | 67. CLD_MaxPhonemeFrequency |
| 31. CLD_C1FrequencyRaw | 68. CLD_C1MaxPhonemeFrequency |
| 32. CLD_C2FrequencyRaw | 69. CLD_C2MaxPhonemeFrequency |
| 33. CLD_FrequencyWeibo | 70. CLD_C1InitialPhonemeFrequency |
| 34. CLD_C1FrequencyWeibo | 71. CLD_C2InitialPhonemeFrequency |
| 35. CLD_C2FrequencyWeibo | 72. CLD_MeanDiphoneFrequency |
| 36. CLD_FrequencyRawWeibo | 73. CLD_MinDiphoneFrequency |
| 37. CLD_C1FrequencyRawWeibo | 74. CLD_MaxDiphoneFrequency |

- | | |
|-------------------------------|--|
| 75. CLD_C1Friends | 84. CLD_C1BackwardEntropy |
| 76. CLD_C2Friends | 85. CLD_PMI |
| 77. CLD_C1HomographTokens | 86. CLD_PSPMI |
| 78. CLD_C2HomographTokens | 87. CLD_TScore |
| 79. CLD_C1FriendsFrequency | 88. CLD_PSTScore |
| 80. CLD_C2FriendsFrequency | 89. CLD_C1ConditionalProbability |
| 81. CLD_C1HomophonesFrequency | 90. CLD_C1BackwardConditionalProbability |
| 82. CLD_C2HomophonesFrequency | 91. CLD_EntropyCharacterFrequencies |
| 83. CLD_C1Entropy | |

S1.2.3 Similarity-based matching

For each homonymous word form, we computed cosine similarity between its normalized 91-feature vector and the vectors of all eligible polysemous and monosemous candidates. Cosine similarity was defined as:

$$\text{cosine}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

where $\mathbf{x}$ is the homonym feature vector and $\mathbf{y}$ is the candidate feature vector. For each homonym, candidates were ranked by similarity and the **top 5** candidates were retained for assignment.

To avoid duplicate assignments, we did not simply assign the top-ranked candidate to each homonym in a fixed order. Instead, we used a **global uniqueness constraint** combined with an iterative **gap-based conflict-resolution procedure**. When two or more homonyms competed for the same candidate, priority was given to the homonym whose currently best available candidate showed the largest drop in similarity relative to its next-best available alternative. Intuitively, this prioritizes assignments for homonyms whose top match is most indispensable. The procedure was repeated until each homonym had a unique assigned polyseme and a unique assigned monoseme. In practice, every homonym received a valid match within the retained candidate range.

S1.2.4 Sense-count alignment

The initial one-to-one headword matches produced closely matched word-level candidates, but not equal **meaning totals** across lexical types. The 125 homonymous word forms contributed **333 meanings**, whereas the initially assigned polysemous and monosemous candidates contributed fewer meanings in total. Because the benchmark was intended for meaning-level comparison, we performed an additional alignment step so that each word type contributed **333 meanings**.

For the **polyseme** benchmark, we preferentially substituted verified **three-sense polysemous candidates** when doing so increased sense-count comparability while preserving close psycholinguistic matching. These substitute candidates were identified in the same normalized 91-feature space used in the original matching. The same global uniqueness principles were applied: candidates already used elsewhere were not reassigned, and conflicts were resolved by the same similarity-gap logic.

For the **monoseme** benchmark, sense-count alignment required additional distinct monosemous word forms because each monosemous word contributes only one meaning. We therefore extended the initial monoseme matches by drawing additional unique monosemous candidates from the ranked similarity lists for each homonym until the monoseme benchmark also contributed **333 meanings**. This yielded the final set of **333 monosemous word forms**.

S1.2.5 Post hoc balance diagnostics

After matching and sense-count alignment, we conducted post hoc balance checks on the **raw (non-normalized)** feature values. Pairwise comparisons between matched sets were performed using Welch's unequal-variance *t* tests across the 91 features, with Benjamini-Hochberg correction for multiple testing. We also ran Wilcoxon rank-sum tests as a robustness check against unequal group sizes and distributional differences. These diagnostics indicated especially close balance for the homonym–polyseme comparison and only limited residual imbalance for a small subset of homonym–monoseme contrasts. Full diagnostic outputs and intermediate matching files are archived in the public repository.

The final Experiment 1 benchmark set therefore consisted of **125 homonymous word forms, 125 polysemous word forms, and 333 monosemous word forms**, contributing **333 meanings per word type**.

S1.3 Meaning-level gloss editing and example-sentence sourcing

Each meaning-level item was presented with three components: (1) the target word form, (2) a POS label and gloss identifying the intended meaning, and (3) an example sentence illustrating that meaning in context. Because dictionary definitions vary in compactness and readability, glosses were lightly edited under fixed rules designed to improve clarity while preserving the original sense distinction.

The gloss-editing rules were as follows:

1. **Rhotic/érhuà tags** were moved from the beginning of a gloss to the end and rewritten as plain-language notes.
2. **Register labels** such as colloquial or literary were expanded from dictionary abbreviations into plain-language notes.
3. **Overly technical gloss material** was trimmed when it was not necessary for identifying the sense, while preserving the definitional core.
4. **Cross-reference-only glosses** (e.g., "see X") were replaced with the corresponding sense description from the referenced entry.
5. **Parenthetical explanations embedded in example sentences** were removed when they were editorial rather than part of the sentence itself.
6. **Parenthesized omitted subjects** were deleted when they were not needed for interpretation.

These edits were intentionally conservative: no new meanings were introduced, and the goal was not to paraphrase freely but to preserve dictionary-based sense distinctions in a format suitable for rating.

Example sentences were sourced according to a fixed hierarchy. When *Modern Chinese Dictionary* already provided an appropriate example sentence for a given meaning, we used that example. When no suitable example was available there, we searched first the *Comprehensive Dictionary of Modern Chinese* (Institute of Linguistics, 2006) and then the *Comprehensive Dictionary of Chinese* (Institute of Linguistics, 2008). If an example was still unavailable, we sourced one from authorized online resources (Oxford Languages or Baidu Education). In a small number of cases, examples in the original dictionary were judged to be clearly incorrect or unnatural, and were replaced using the same source hierarchy. To maximize naturalness and reproducibility, example sentences were drawn from authoritative sources rather than authored by the researchers. The source of every example sentence is recorded in the released database.

S1.4 Full participant instructions

S1.4.1 Original Chinese instructions

您好！欢迎参加我们的实验。

在汉语中，有些词语具有多个意思，例如"上任"既可以是动词，指官吏就职，如"走马上任"；也可以是名词，指前一任官吏，如"张其昌已被选为这个乡的乡长，且看他如何收拾上任留下的烂摊子"。我们分别在不同的时间学会这些不同的意思，有的早一些学会，有的晚一些。

本次实验要求您根据自己的语感判断一个词语的某个意思是在什么时候学会的，1分到7分打分。0到2岁给1分，2到4岁给2分，依次类推，13岁以后才学会这个词语的这个意思给7分。若您不知道这个词语的这个意思，请直接给7分。

实验开始时会出现"+"注视点提醒您准备开始实验，接下来会出现需要判断的词语，在该词语下会出现其某个意思及对应词性，在其下会出现对应意思的例句提示您这个意思的用法。请您评判这个词语的这个意思是在什么年龄段学会的，并快速准确地打分。

如果您准备好了，可以按空格键开始练习。

S1.4.2 English translation

Hello, and welcome to our experiment.

In Chinese, some words have multiple meanings. For example, 上任 (*shàng rèn*) can be used as a verb meaning "to take up an official post," as in 走马上任 (*zǒu mǎ shàng rèn*; "to take up a post, arriving promptly to assume office"); it can also be used as a noun meaning "the previous officeholder," as in the sentence "张其昌已被选为这个乡的乡长，且看他如何收拾上任留下的烂摊子" ("Zhang Qichang has been elected as the head of this township; let us see how he cleans up the mess left by his predecessor").

We learn these different meanings at different times—some earlier and some later.

In this experiment, you will use your own language intuition to judge when you learned a particular meaning of a word, and give a rating from 1 to 7. Give **1** for ages **0–2**, **2** for ages **2–4**, and so on; give **7** if you learned that meaning after age 13. If you do not know that meaning of the word, please select **7**.

At the beginning of the experiment, a "+" fixation point will appear to remind you to get ready. Next, the word to be judged will appear. Below the word, you will see one of its meanings and the corresponding part of speech; below that, you will see an example sentence showing how the meaning is used. Please judge the age range at which you learned that meaning, and respond as quickly and accurately as possible.

If you are ready, you can press the space bar to start the practice.

S2. Supplementary statistical tables

Table S1

Fixed effects from the linear mixed-effects model predicting logRT in Experiment 1

Predictor	<i>b</i>	SE	<i>df</i>	<i>t</i>	<i>p</i>	95%
Intercept	7.887	0.039	53.6	200.50	< .001	[7.80
Within-subject RESP (RESP _w)	0.069	0.007	71.5	10.34	< .001	[0.0
Word type: polyseme (vs. homonym)	-0.044	0.009	977.3	-5.09	< .001	[-0.0
Word type: monoseme (vs. homonym)	-0.114	0.009	976.5	-13.26	< .001	[-0.1
Trial order (trial _z)	-0.115	0.002	54,950	-63.70	< .001	[-0.1
Between-subject RESP (RESP _b)	0.144	0.037	57.0	3.86	< .001	[0.0
RESP _w × polyseme	0.005	0.004	23,650	1.21	.228	[-0.0
RESP _w × monoseme	-0.025	0.004	20,540	-6.44	< .001	[-0.0

Note. Model formula: $\text{logRT} \sim \text{RESP}_w * \text{word_type} + \text{trial_z} + \text{RESP}_b + (1 + \text{RESP}_w | \text{Subject}) + (1 | \text{item_id})$. The reference level for *word_type* is homonym. RESP_w is the within-

subject centered RESP score; $RESP_b$ is the z-scored subject mean RESP; $trial_z$ is z-scored trial order. Confidence intervals are Wald 95% confidence intervals. Positive coefficients indicate longer logRT.

Table S2

Random effects for the linear mixed-effects model predicting logRT in Experiment 1

Grouping factor	Random effect	Variance	SD	Correlation
Item	Intercept	0.0092	0.096	—
Subject	Intercept	0.0890	0.298	—
Subject	Slope for $RESP_w$	0.0022	0.047	0.30
Residual	—	0.1672	0.409	—

Note. Number of observations = 55,346; subjects = 59; items = 999. The reported correlation is the correlation between the subject-specific random intercept and the subject-specific random slope for $RESP_w$.

Table S3

Simple slopes of within-subject RESP by word type and pairwise comparisons

A. Simple slopes

Word type	RESP slope	SE	95% CI
Homonym	0.0692	0.0067	[0.0561, 0.0823]
Polyseme	0.0738	0.0068	[0.0604, 0.0871]
Monoseme	0.0438	0.0069	[0.0303, 0.0573]

B. Pairwise comparisons of slopes

Contrast	Estimate	SE	z	p (Holm-adjusted)
Homonym – polyseme	-0.0046	0.0038	-1.21	.228
Homonym – monoseme	0.0254	0.0039	6.44	< .001
Polyseme – monoseme	0.0300	0.0041	7.25	< .001

Note. Simple slopes were estimated using `emtrends(m_rt, ~ word_type, var = "RESP_w")`.

Pairwise comparisons were computed with Holm correction for multiple testing. The `emtrends` output used the asymptotic method ($df = \infty$).

Table S4

Fixed and random effects for the linear mixed-effects model predicting logRT in Experiment 2

A. Fixed effects

Predictor	b	SE	df	t	p	95%
Intercept	7.959	0.036	97.810	223.044	< .001	[7.88

Predictor	<i>b</i>	SE	<i>df</i>	<i>t</i>	<i>p</i>	95%
Within-subject RESP (RESP _w)	0.000	0.007	101.000	0.037	.971	[-0.0
Trial order (trial _z)	-0.240	0.004	50,690.000	-60.221	< .001	[-0.2
Between-subject RESP (RESP _b)	-0.011	0.059	96.480	-0.190	.850	[-0.1

B. Random effects

Grouping factor	Random effect	Variance	SD	Correlation
Item	Intercept	0.013	0.113	—
Subject	Intercept	0.125	0.353	—
Subject	Slope for RESP _w	0.005	0.070	-0.050
Residual	—	0.202	0.450	—

Note. Model formula: $\log RT \sim \text{RESP}_w + \text{trial}_z + \text{RESP}_b + (1 + \text{RESP}_w \mid \text{Subject}) + (1 \mid \text{item_id})$. Estimation used REML with Satterthwaite degrees of freedom (`lmerTest`). Number of observations = 52,792; participants = 99; items = 1,108. LogRT was trimmed within participant using the ± 2.5 MAD criterion on logRT. Confidence intervals are Wald 95% confidence intervals. The RESP_w point estimate is very small (0.00027) and therefore rounds to 0.000 at three decimals. The reported correlation is the correlation between the subject-specific random intercept and the subject-specific random slope for RESP_w.

Table S5

Fixed effects for the predictive-validity mixed-effects model predicting log RT in the sentence-based homonym disambiguation task

Predictor	<i>b</i>	SE	<i>df</i>	<i>t</i>	
Intercept	8.261	0.034	66.973	244.957	< .0
Sense	-0.044	0.013	144.237	-3.475	< .0
Context	0.012	0.008	494.797	1.435	.1
POSWord	-0.018	0.008	493.892	-2.139	.0
Trial _z	-0.121	0.003	25,212.859	-47.830	< .0
Word_logW _z	0.023	0.010	490.053	2.175	.0
AoA _{word,z}	0.026	0.010	498.259	2.578	.0
AoA _{mean,z}	0.033	0.011	497.063	2.925	.0
Sense × Context	-0.063	0.008	494.817	-7.697	< .0
Sense × POSWord	0.004	0.008	494.679	0.433	.0
Context × POSWord	-0.015	0.008	494.564	-1.798	.0
Sense × Context × POSWord	0.007	0.008	494.572	0.837	.4

Note. Model formula: $\log RT \sim \text{Sense} + \text{Context} + \text{POSWord} + \text{Trial}_z + \text{Word_logW}_z + (1 + \text{Sense} \mid \text{Subject}) + (1 \mid \text{SentencePair_id}) + \text{AoA_word}_z + \text{AoA_mean}_z + \text{Sense}:\text{Context} + \text{Sense}:\text{POSWord} + \text{Context}:\text{POSWord} + \text{Sense}:\text{Context}:\text{POSWord}$. Estimation used Satterthwaite degrees of freedom (`lmerTest`). Analyses were restricted to correct trials. Number of observations = 25,730; sentence pairs = 512; participants = 60. Binary predictors (`Sense` , `Context` , `POSWord`) were sum-coded. Predictors with suffix $_z$ were z-scored. Confidence intervals are Wald 95% confidence intervals.

Table S6

Random effects for the predictive-validity mixed-effects model predicting log RT in the sentence-based homonym disambiguation task

Grouping factor	Random effect	Variance	SD	Correlation
SentencePair _{id}	Intercept	0.031	0.177	—
Subject	Intercept	0.064	0.253	—
Subject	Slope for Sense	0.006	0.075	0.431
Residual	—	0.159	0.399	—

Note. Number of observations = 25,730; sentence pairs = 512; participants = 60. The reported correlation is the correlation between the subject-specific random intercept and the subject-specific random slope for `Sense` .

Table S7

Nested model comparisons for the predictive-validity model-building sequence

Model	Description	<i>npar</i>	AIC	BIC	logLi
M1	Base structural model	15	27,464.76	27,587.10	-13,717.3
M2	M1 + word-level AoA	16	27,448.82	27,579.31	-13,708.4
M3	M2 + mean meaning-level AoA	17	27,442.34	27,580.98	-13,704.1
M4	M3 + between-meaning AoA difference	18	27,444.00	27,590.80	-13,704.0

Note. Models were compared using likelihood-ratio tests for nested mixed models fit by maximum likelihood. The base structural model included `Sense` , `Context` , `POSWord` , all lower- and higher-order interactions among those factors, `Trial_z` , `Word_logW_z` , random intercepts for subjects and sentence pairs, and a by-subject random slope for `Sense` . M2 added z-scored word-level AoA; M3 added z-scored mean meaning-level AoA; M4 added z-scored absolute difference between the two meaning-level AoA values. Analyses were restricted to correct trials. *p*-values correspond to the χ^2 test.